Reference-Free Selective Spectral Correction for Frequency-Domain Backdoor Mitigation

Project Phase 2

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Guide Approval Mail

1. Introduction and Motivation

Why Backdoor Security Matters in Image Classification

- Deep-learning classifiers are increasingly used in safety-sensitive applications.
- A backdoor can keep normal accuracy on clean images while causing targeted errors when a trigger is present.
- A defense should remove trigger dependence without destroying useful image information.

clean image -> classifier -> correct class
triggered image -> classifier -> attacker target class

- Frequency-domain triggers can be visually subtle and difficult to identify directly in pixel space.
- Current frequency-domain defenses suppress all frequency regions — including both trigger frequencies and useful signal frequencies. This is a blunt instrument that degrades legitimate model performance.
- **“how can we weaken a learned trigger dependency without removing the natural frequency information that the classifier needs?”**

2. Problem Statement

Problem statement:

- To develop a selective and adaptive frequency-domain defense that suppresses trigger-related spectral information while maintaining clean-image classification performance.

Objectives

- Identify suspicious amplitude changes caused by a trigger.
- Learn an adaptive correction map rather than applying one global filter.
- Repair the suspicious classifier so it no longer depends on the trigger.
- Evaluate the approach across resolutions, trigger strengths, trigger functions and unknown images.

3. Literature Survey and Research Gap

Source	What the paper does	Limitation for our problem	What our project improves
Freq-Pret	Retrains the classifier using frequency-perturbed images.	Changes model weights and requires a clean subset plus retraining.	Keeps the classifier frozen and produces a corrected image from one input.
BAM	Learns an input purifier from simulated backdoor behavior.	General image-space purification; spectral support is less explicit.	Uses amplitude, phase, residual and frequency position, with causal support tests.
Semantic-aware	Uses a trusted model to detect suspicious behavior, then performs unlearning.	Needs a trusted comparison model and modifies the suspicious classifier.	Uses no second model at inference and leaves the classifier unchanged.

Research gap: protect a frozen third-party classifier by correcting one image in frequency space, without requiring its matching clean image at inference.

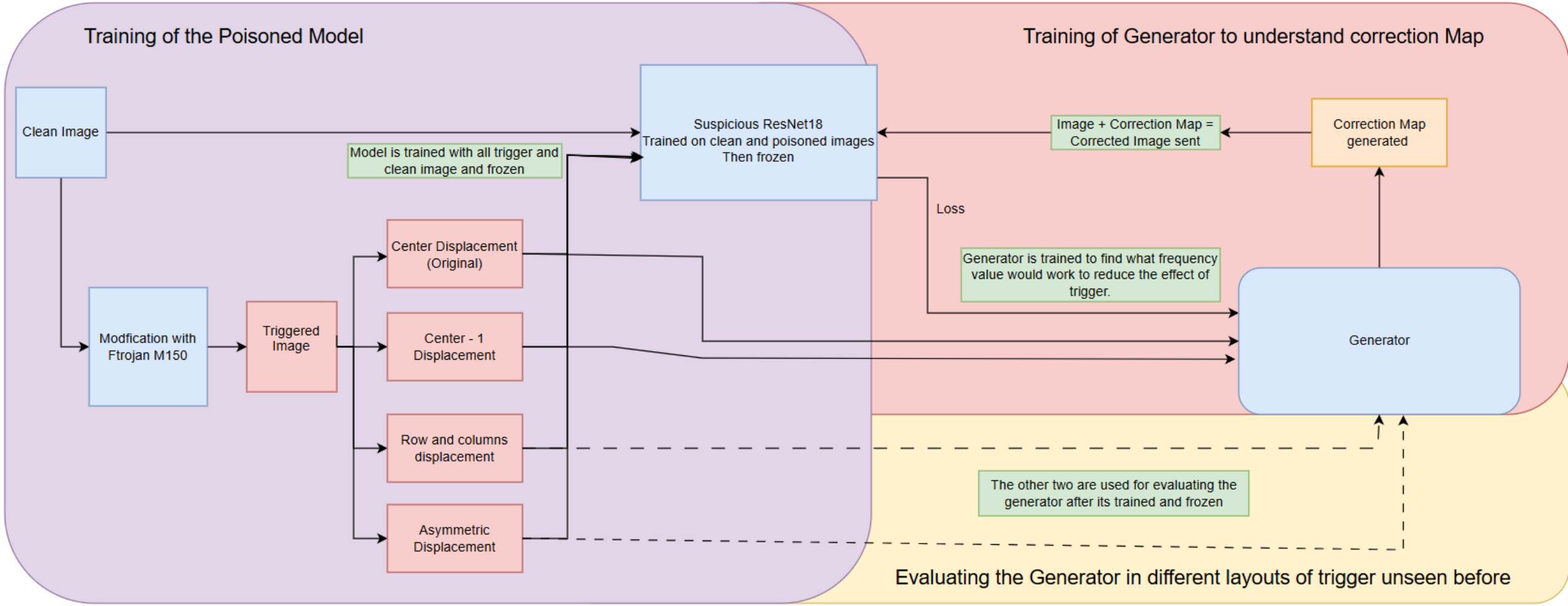
3.1 Where the First Panel Left the Proiect

What Phase 1 established	What remained unresolved
Complete poisoning, classifier, generator and repair pipeline	Generator received clean and triggered spectral information
CIFAR-100, Tiny ImageNet and STL-10	Absolute difference made trigger evidence directly visible
32x32, 64x64 and 96x96 resolutions	Images and triggers were relatively controlled
Strength, function and FIBA-style experiments	Unknown trigger and noise behaviour remained unclear

3.2 Questions Raised After the First Panel

Panel questions	Answer
What if the clean image is unavailable?	The new generator needs only the incoming trigger image. It does not need the original clean version while correcting it.
Is absolute subtraction doing the work?	During training it sees many examples and learns which frequency patterns repeatedly lead to the attack. For a new image, it predicts only the spectral changes that appear suspicious.
Can it handle complex texture images?	Tested on 224x224 DTD with 47 texture classes.
What happens if the attacker changes the frequency location?	We hid two active trigger positions from generator training. On those positions, attack success fell from 99.40% to 7.20% on average.
Does the generator remove the trigger, or does it simply remove any noise?	It did not blindly erase all frequencies. Selective correction worked much better than broad filtering, while strong unrelated noise revealed a clear failure boundary.

4. Proposed Methodology

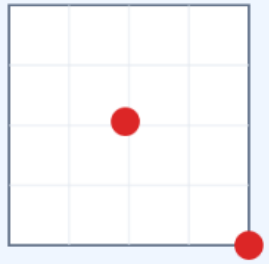


4. Proposed Methodology

Four coefficient layouts used by the final active attacker

Dots are DCT coefficient coordinates inside a 32 x 32 block, not image pixel locations

Original - seen by generator

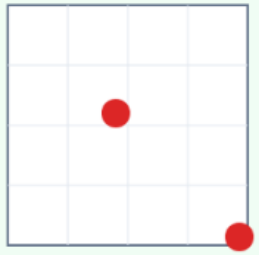


(15,15), (31,31)

Principal calibrated location.

Learns middle/high-frequency repeating chroma evidence.

Near minus 1 - seen

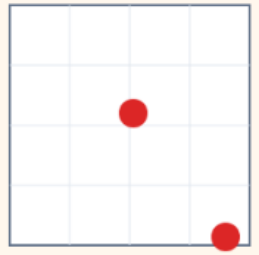


(14,14), (30,30)

Small diagonal displacement.

Adds position diversity during generator training.

Near mixed - held out

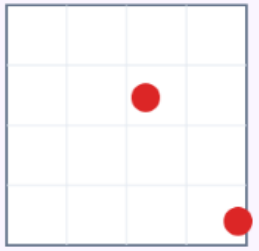


(14,16), (30,28)

Rows and columns move differently.

Tests unseen spectral position without changing trigger family.

Middle shift - held out



(12,18), (28,30)

Larger asymmetric displacement.

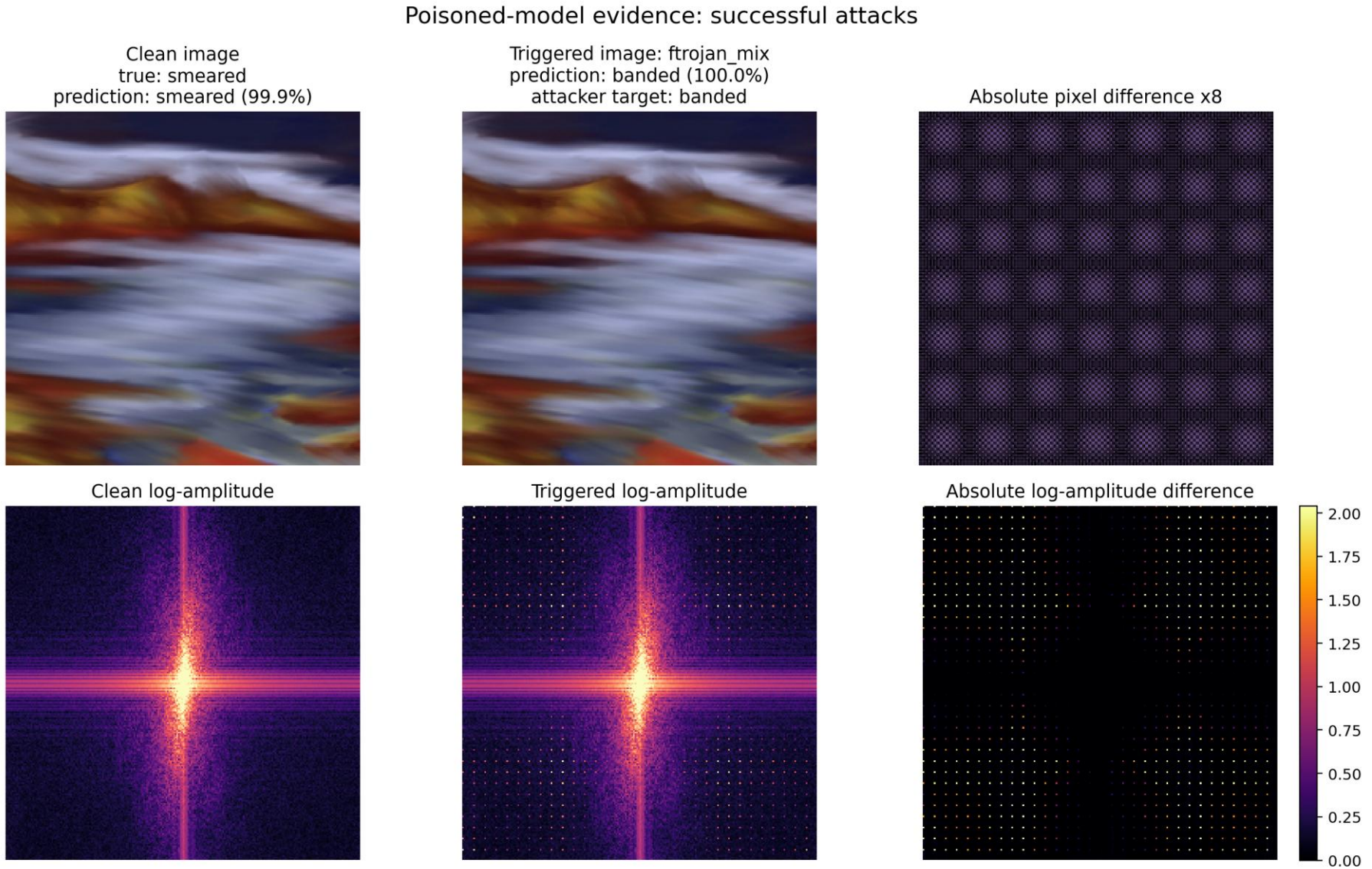
Second active location hidden from generator training.

The classifier was trained on all four layouts; the generator was trained on the two blue/green layouts and evaluated on the two orange/purple layouts.

4. Proposed Methodology

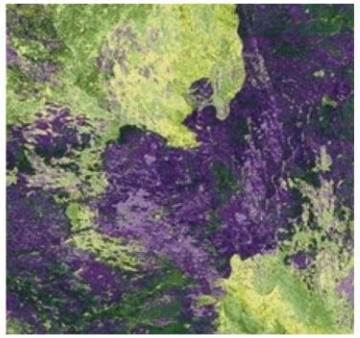
- Start with a clean "striped" image.
- Add Trigger Layout
  - The suspicious ResNet18 now predicts "banded“.
  - Give only the triggered image to the generator.
  - The generator examines its frequency representation.
  - It predicts:
    - Which frequency regions should be changed?
    - Should their amplitude increase or decrease?
    - How large should the change be?
- Reconstruct the corrected image.
- Give the corrected image to the frozen ResNet18.
- Compare the result with the known original label "striped".
- If ResNet18 still predicts "banded", update the generator.
- If it predicts "striped" while the image remains visually similar, reward that generator behaviour.

4. Proposed Methodology

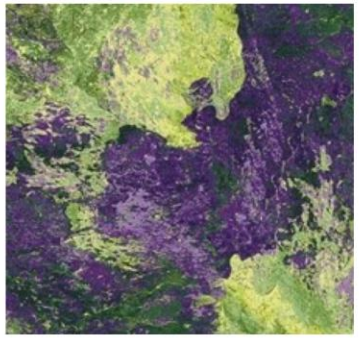


How the reference-free generator corrected one successful FTrojan attack
true class: blotchy | generator input: triggered image only

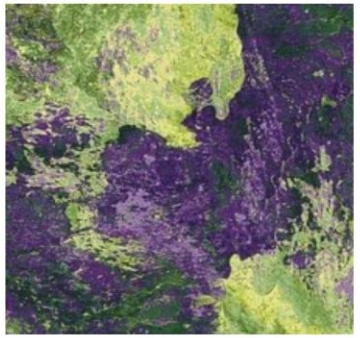
clean
pred: blotchy



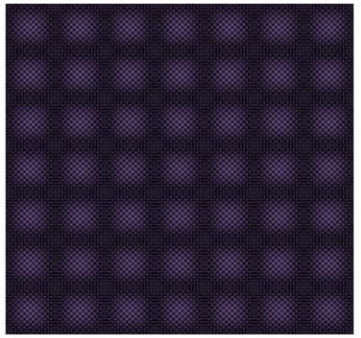
triggered
pred: banded



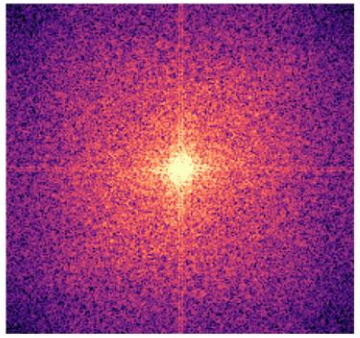
corrected
pred: blotchy



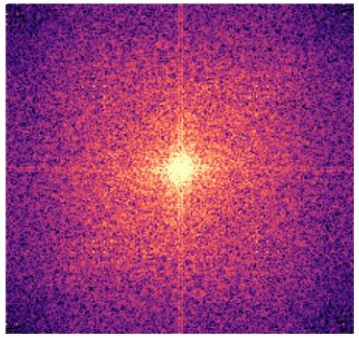
pixel difference |x8|



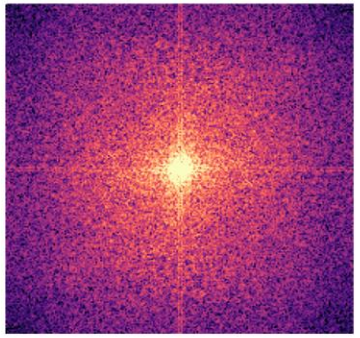
clean log amplitude
(shared scale)



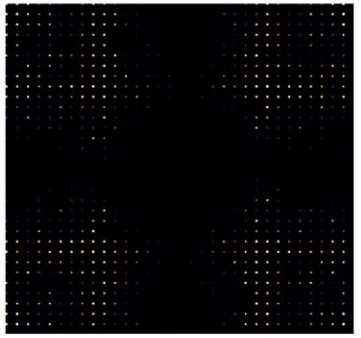
trigger log amplitude
(shared scale)



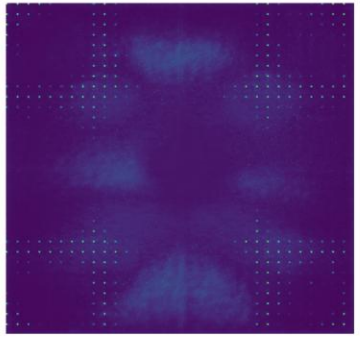
corrected log amplitude
(shared scale)



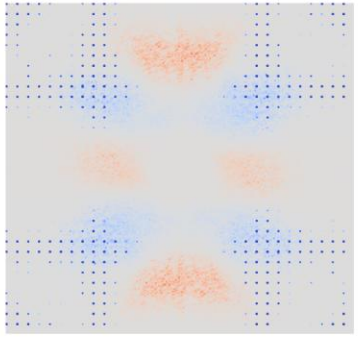
|trigger - clean| spectrum
(enhanced)



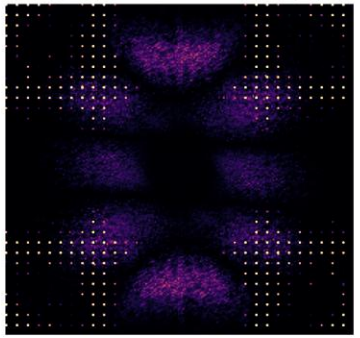
predicted gate M



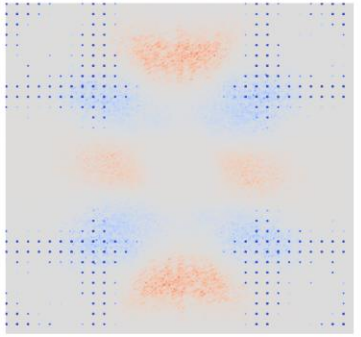
generator signed correction



|effective correction|



corrected - trigger spectrum



Pixel MAE trigger->clean 0.02023; corrected->clean 0.00575 | Log-amplitude MAE trigger->clean 0.01039; corrected->clean 0.01835

4. Proposed Methodology

Algorithm 1 Reference-Free Selective Spectral Correction

Require: clean labeled set D; target class y*; FTrojan-style trigger T; seen positions P_seen; loss weights.

Ensure: suspicious classifier f_s; generator G_theta; corrected prediction and evaluation metrics.

One-line summary: verify an active attack -> learn from one-image spectral evidence -> reconstruct -> classify.

Stage A - Train and verify the suspicious classifier

Select non-target images and append poisoned pairs (T(x,p), y*) to retained clean rows.
Fine-tune the pretrained classifier on clean and poisoned rows.
Verify clean accuracy, ASR lift, and conditional ASR; freeze the validated model as f_s.

Stage B - Train the reference-free generator

for each generator epoch and clean minibatch (x,y) **do**
  t <- T(x,p), where p is sampled from P_seen.
  Only t enters the generator. Clean x supplies a target for the losses, not a second inference input.
  (L_t, P_t) <- LogAmplitudePhase(t).
  Represent how much frequency energy exists and retain where its structure is arranged.
  Z <- Evidence15(L_t, P_t).
  Z contains normalized amplitude, local residual, sin/cos phase, and frequency coordinates.
  (M, Delta) <- G_theta(Z); E <- M x Delta.
  M controls where intervention is allowed; Delta controls direction and size.
  L_corr <- max(L_t + E, 0).
  x_corr <- clip(Re[IFFT((exp(L_corr)-1) exp(j P_t))], 0, 1).
  Change log amplitude selectively, preserve the input phase, and reconstruct a valid RGB image.
  x_id <- Correct(x; G_theta).
  L <- L_cls + 4 L_img + L_spec + 2 L_id + 0.02 L_sp + 0.01 L_sm.
  Reward label recovery and reconstruction; penalize needless, dense, or irregular correction.
  Update only theta; keep the suspicious classifier frozen.
end for

Stage C - Correct one possibly suspicious image at inference

Receive one image u; compute Z(u) from u alone.
Predict E(u), correct amplitude, preserve u's phase, and reconstruct u_corr.
Return f_s(u_corr), corrected image, and optional correction evidence.
Report clean accuracy, ASR, true-label recovery, ablations, baselines, and TNR behavior.

Deployment rule: the clean counterpart and true label are absent; the trained weights process one input once.

5. Datasets

Property	Value
Texture classes	47
Images per class	120
Total source images	5,640
Official split-1 training	1,880
Official split-1 validation	1,880
Locked test	1,880
Evaluation resolution	224x224 RGB
Classifier	ImageNet-pretrained ResNet18
Attacker target	Class 0, banded

6. Implementation Progress and Discussion

Candidate	Evidence	Decision
Direct Fourier-band trigger	Triggered target 13.53%; clean target 9.89%	Rejected due to weak ASR
Haar-LH wavelet	ASR 9.67%; clean target 9.51%; lift 0.16 pp	Rejected due to weak ASR
Early FTrojan M50	Poison-row target 95.21%; validation approximately 15.9%	Rejected due to weak validation phase
FTrojan M100	ASR 86.63%; clean target 5.60%; lift 81.03 pp	Accepted
FTrojan M150, four positions	Every position approximately 99.4% to 99.7% ASR	Accepted and Final benchmark

6. Implementation Progress and Discussion

Component	Final advanced configuration
Dataset	DTD, official split 1
Input	224x224 RGB crop
Classifier	ImageNet-pretrained ResNet18, 47 classes
Attack	FTrojan-style block-DCT, four positions, magnitude 150
Target	banded
Generator	771,238-parameter encoder-decoder
Generator input	Single-image 15-channel FFT evidence
Generator output	Gate plus bounded signed log-amplitude proposal
Phase policy	Preserve incoming phase
Evaluation	1,840 non-target validation sources per position

7. Standard Algorithms and Evaluation Metrics

Core algorithms

FFT / IFFT
- Decompose image into amplitude and phase.

ResNet-18
- ImageNet-pretrained 47-class texture classifier.

Correction generator
- Predicts a gate and signed log-amplitude change.

Optimization
- Classification, image, spectral, identity, sparsity and smoothness losses.

Evaluation metrics

Clean accuracy
- Normal classification utility.

Attack Success Rate (ASR)
- Fraction forced into the attacker's target class.

True-label recovery
- Correct classification after correction.

Target-rate lift and conditional ASR
- Verify that the trigger caused new failures.

Pixel L1 and TNR
- Measure image change and trigger/noise stress.

8. Tools and Implementation Expertise

Training and models

Python and PyTorch
- Dataset pipelines, poisoning, model training and generator optimization.

Kaggle GPU
- NVIDIA T4 sessions for attack and generator experiments.

Pretrained learning
- ImageNet ResNet-18 adapted to 47 DTD classes.

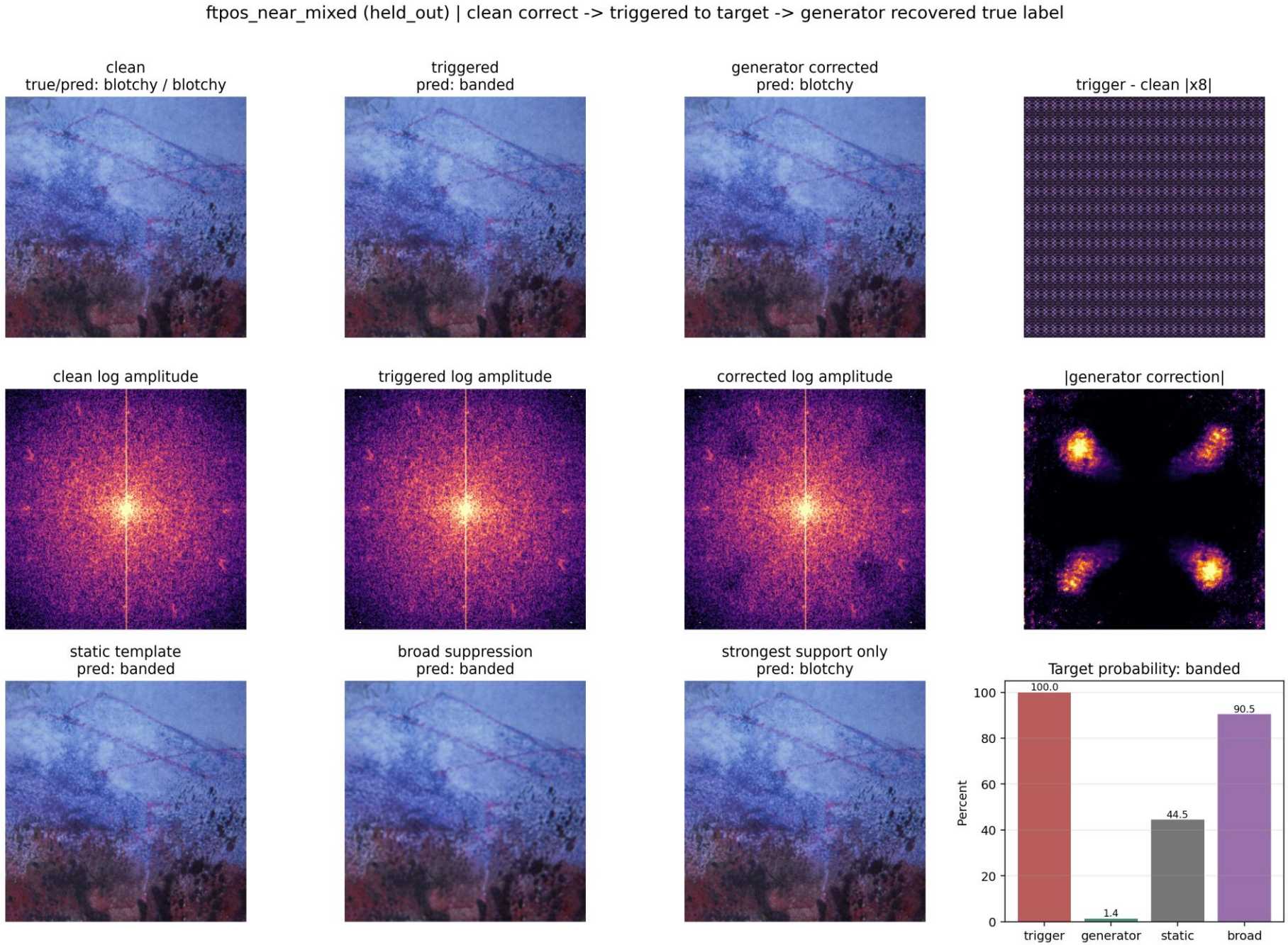
Analysis and evidence

Signal processing
- FFT/IFFT, log amplitude, phase preservation and block-DCT trigger injection.

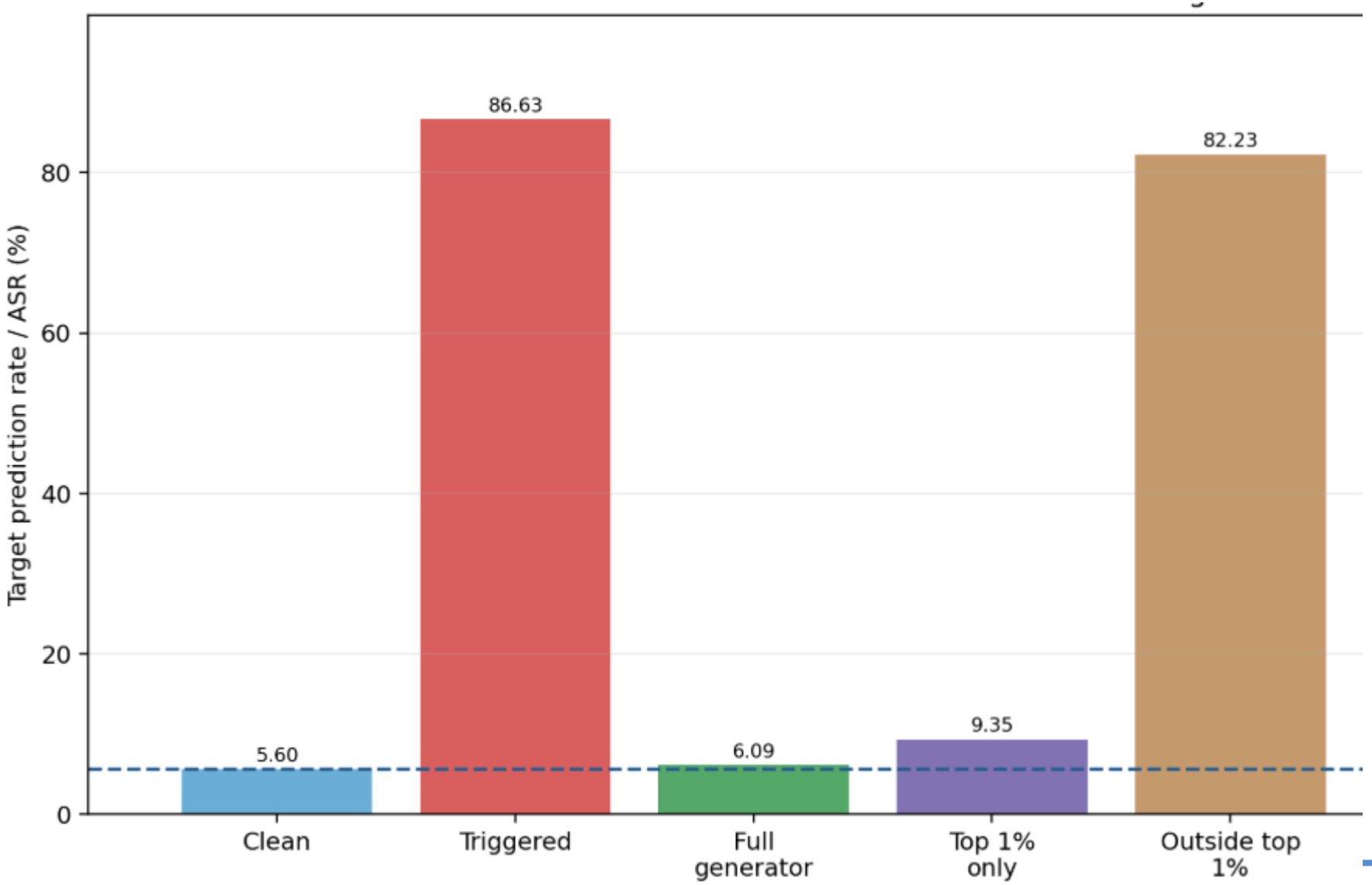
Reproducibility
- Saved checkpoints, JSON summaries, manifests and fixed evaluation splits.

Visualization
- Correction maps, spectral panels, ablation charts and a CPU demonstration notebook.

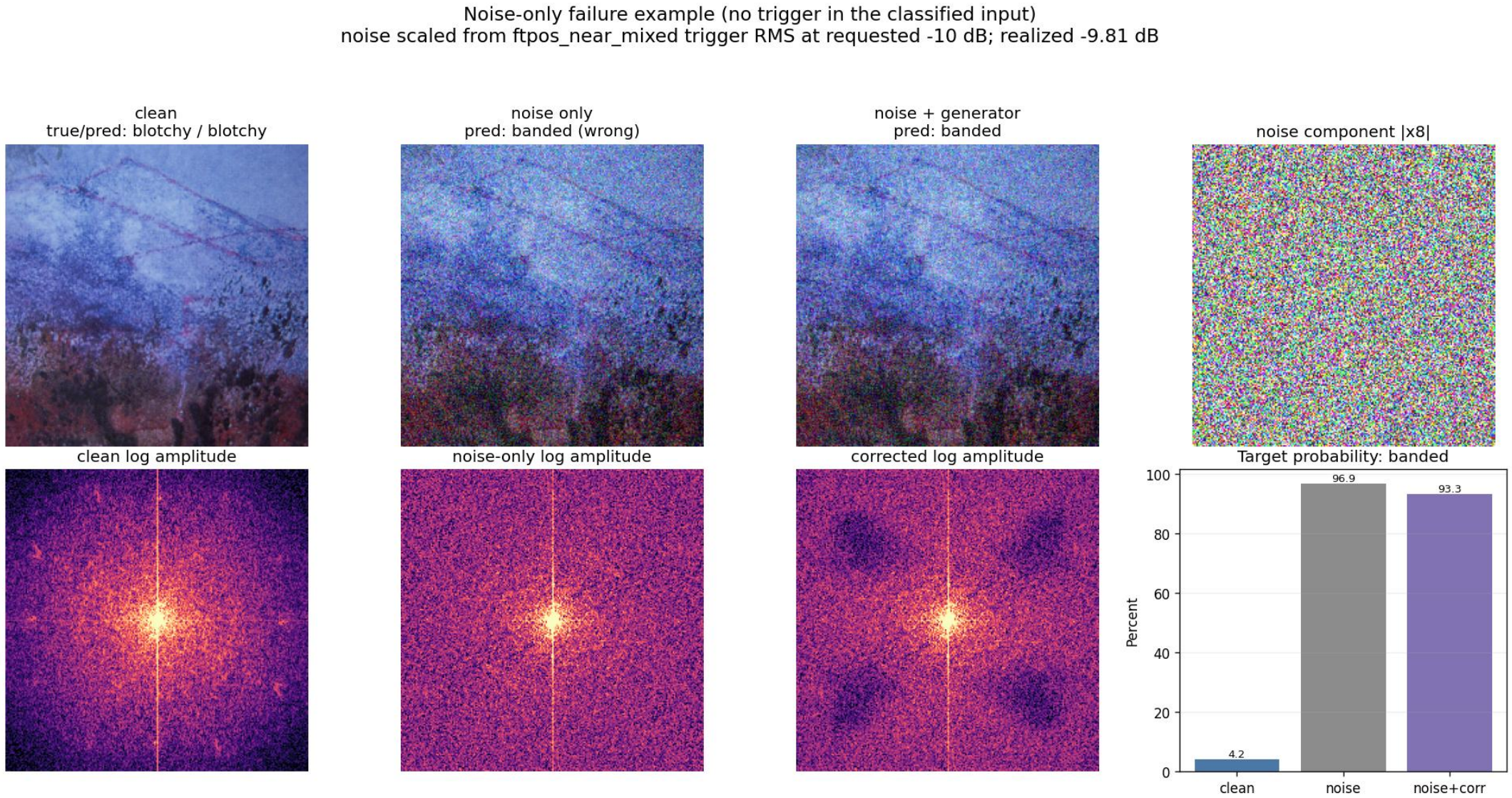
9. Results


ftpos_near_mixed (held_out) | clean correct -> triggered to target -> generator recovered true label
clean
true/pred: blotchy / blotchy
triggered
pred: banded
generator corrected
pred: blotchy
trigger - clean |x8|
clean log amplitude
triggered log amplitude
corrected log amplitude
|generator correction|
static template
pred: banded
broad suppression
pred: banded
strongest support only
pred: blotchy
Target probability: banded
Percent
100.0
1.4
44.5
90.5
trigger
generator
static
broad

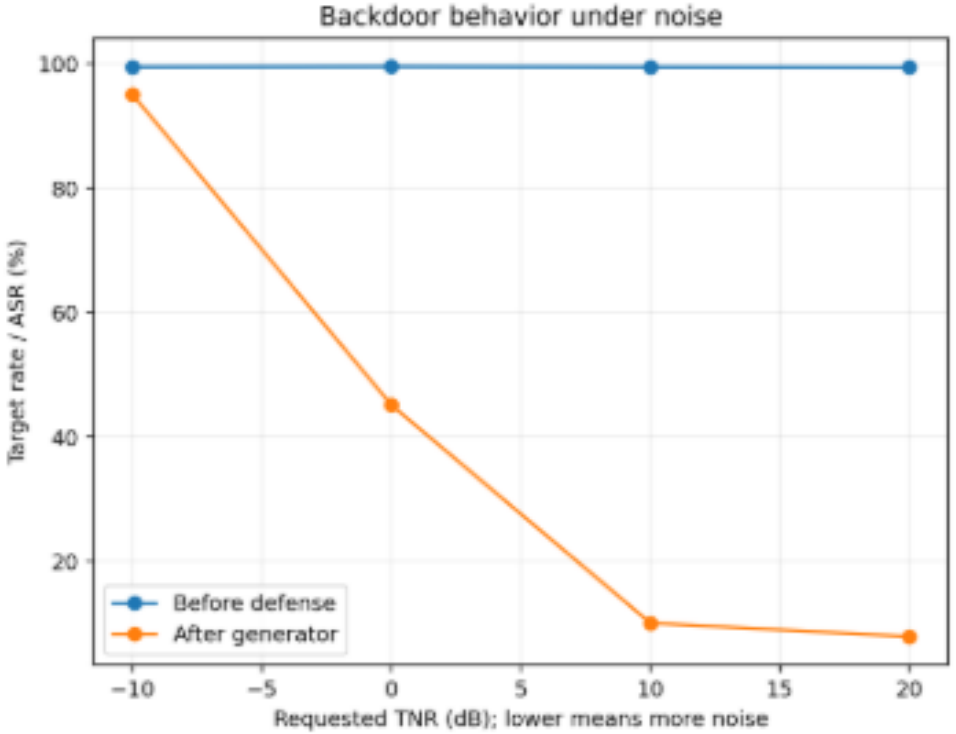
9. Results

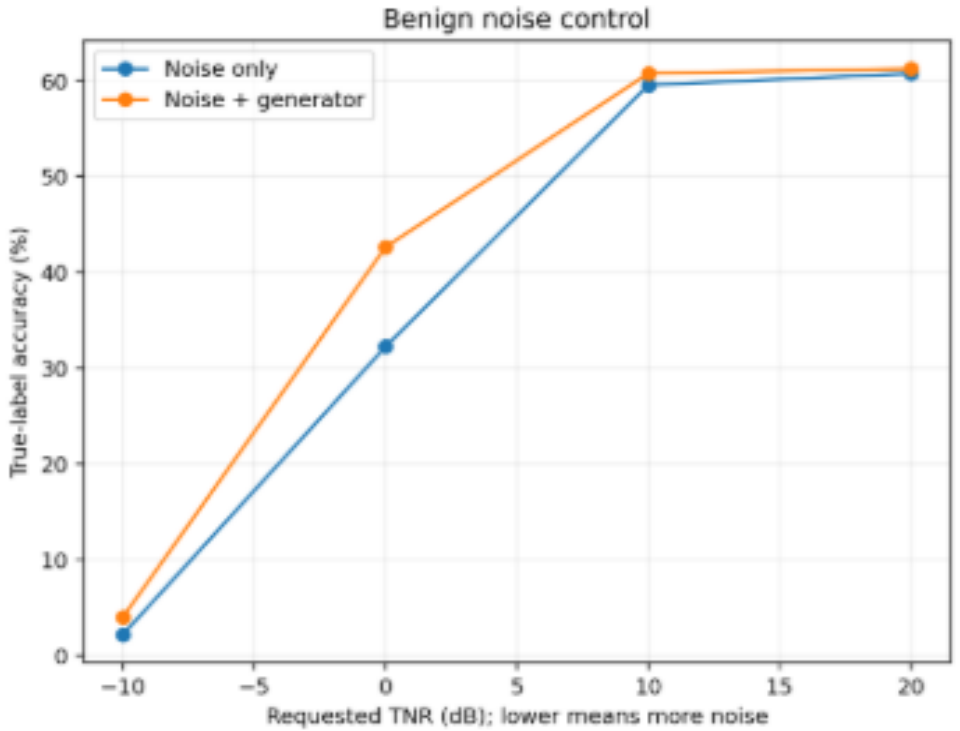

Target prediction rate / ASR (%)
0
20
40
60
80
5.60
86.63
6.09
9.35
82.23
Clean
Triggered
Full generator
Top 1% only
Outside top 1%

9. Results - Trigger Versus Noise



9. Results - Trigger Versus Noise





Condition	Corrected ASR	Corrected accuracy
Trigger only	7.64%	60.29%
+20 dB, trigger much stronger than noise	7.85%	60.19%
+10 dB, trigger stronger than noise	10.04%	58.93%
0 dB, equal RMS power	45.11%	35.52%
-10 dB, noise stronger than trigger	95.05%	3.48%

10. Demonstration

```python
from pathlib import Path
import json
import os
import subprocess
import sys

import torch
from IPython.display import display
from PIL import Image

def find_repo_root(start):
    start = Path(start).resolve()
    for candidate in (start, *start.parents):
        if (candidate / 'scripts' / 'demo_dtd_reference_free_inference.py').is_file():
            return candidate
    raise FileNotFoundError('Could not locate the Research_Paper repository root.')

REPO_ROOT = find_repo_root(Path.cwd())
os.chdir(REPO_ROOT)
if str(REPO_ROOT) not in sys.path:
    sys.path.insert(0, str(REPO_ROOT))

DEVICE = 'cpu'
print('Repository:', REPO_ROOT)
print('Python:', sys.version.split()[0])
print('PyTorch:', torch.__version__)
print('Selected device:', DEVICE)
```

Python

```python
CLASSIFIER_CKPT = REPO_ROOT / (
    'Advanced_Experiments/dtd_position_attack_selected_v1/'
    'dtd_position_attack_selected_v1/suspicious_classifier_best_attack.pt'
)
GENERATOR_CKPT = REPO_ROOT / (
    'Advanced_Experiments/dtd_position_generalizing_generator_v1/'
    'reference_free_generator_best.pt'
)

for label, path in (
    ('Classifier', CLASSIFIER_CKPT),
    ('Generator', GENERATOR_CKPT),
```

Python

11. SDG Mapping

Potential application relevance: SDG 3

Good Health and Well-being

Medical imaging is a future application area where model backdoors could create serious risks. The current presentation does not claim completed medical validation; SDG 3 is presented as an application motivation and future direction.

Primary mapping: SDG 9

Industry, Innovation and Infrastructure

The project develops a security-oriented machine-learning method intended to make AI systems more reliable and trustworthy, particularly when models may be trained using externally supplied or compromised data.

12. Project Plan for Semester 3

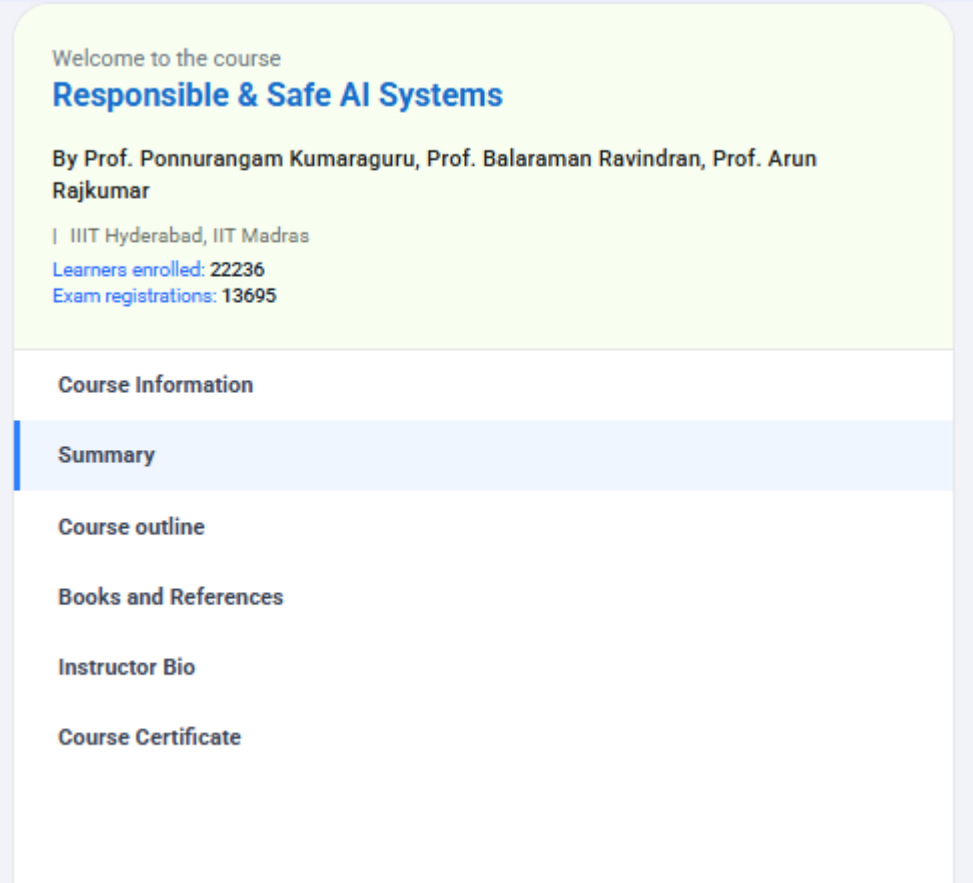
Period	Activity	Status	Expected output
Completed	Attack calibration, generator training, held-out tests, baselines and TNR study	**Done**	Frozen results, checkpoints and evidence figures
Oct-Nov 2026	Organize literature comparison and write IEEE manuscript sections	**In progress**	Complete first paper draft
Nov 2026	Run one locked repeat and multi-seed summary where compute permits	**Planned**	Stronger reproducibility statement
Nov-Dec 2026	Guide review, revisions, formatting and final submission preparation	**Planned**	Submission-ready paper and dissertation material

Primary Phase 3 goal: convert the completed experimental evidence into a defensible, reproducible IEEE-format paper.

13. Domain Expertise Course

Safety and Responsibility AI

- Explains about Trigger and Backdoor attacks in ML



14. Technical Enrichment Activities

Deliverable	Description	Expected deliverables
Submission-ready Research Paper Draft	Prepare a complete research paper in IEEE, Elsevier, Springer, ACM, or any other appropriate conference/journal format based on the project work.	Complete manuscript; Figures and tables; References; Similarity report
Open-source Technical Repository	Maintain a professional GitHub repository containing reusable project resources.	Source code; Well-structured project organization; README; Installation guide; Documentation; Regular commits with version history
Working demonstration	Develop a working end-to-end prototype that demonstrates the core functionality of the project and validates key performance outcomes.	Demonstrate a working model of the project - hardware or deployed software. Also, submit a run-based demo: short video + instructions + the commands/script that regenerates the metrics and sample outputs (with model/config versions).

15. Conclusion

- **Advanced method:** One-image, reference-free selective log-amplitude correction with phase-preserving reconstruction.
- **Main result:** Four-position mean ASR reduced from **99.48% to 7.64%**, with **60.29%** corrected accuracy versus approximately **61.44%** clean accuracy.
- **Evidence quality:** Causal support tests, static/broad baselines, held-out active positions, benign corruptions, and TNR stress testing.
- **Boundary:** Demonstrated on DTD and within one calibrated attack family; cross-family, adaptive, locked-test, and multi-seed evidence remain Phase 3.

16. References

[1] M. Cimpoi et al., "Describing Textures in the Wild," CVPR, 2014.

[2] K. He et al., "Deep Residual Learning for Image Recognition," CVPR, 2016.

[3] T. Wang et al., "An Invisible Black-box Backdoor Attack through Frequency Domain," ECCV, 2022.

[4] P. Zhang, H. Yue, and C. Yuen, "Backdoor Defense Strategy for Image Classification Tasks Based on Frequency Domain Perturbation," IEEE IoT Journal, 2026.

[5] R. Lu et al., "BAM: A Backdoor-Agnostic Input Purification Method," Pattern Recognition, 2026.

[6] K. Fu and J. Dai, "Semantic-Aware Framework for Backdoor Detection in AI Models," IJSWIS, 2025.

[7] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain," 2017.